RetailIQ: An End-to-End E-Commerce Customer and Sales Analytics Platform

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***Abstract* — E-commerce platforms generate large volumes of customer, order, product, payment, return, and review data, making it difficult to obtain a unified view of business performance and customer behavior. This paper presents RetailIQ, an end-to-end e-commerce analytics platform designed to integrate data preprocessing, exploratory data analysis, customer analysis, product performance analysis, sales and revenue analysis, customer segmentation, and interactive visualization. The system uses an integrated Indian e-commerce dataset containing customer, order, order-item, product, payment, return, and review information. Data preprocessing is performed to handle missing values, standardize data types, and prepare the datasets for downstream analysis. Feature engineering is then applied to derive temporal, revenue, profit, discount, customer, and inventory-related attributes. Customer purchasing behavior is analyzed using Recency, Frequency, and Monetary (RFM) measures, followed by customer segmentation using clustering techniques. Product, sales, revenue, payment, return, review, and geographical patterns are analyzed to identify business-relevant trends. The resulting analytical outputs are presented through an interactive dashboard that provides a centralized view of key performance indicators and customer and product insights. The platform demonstrates how integrated data analytics and visualization can support systematic interpretation of e-commerce data and facilitate data-driven business decision-making.**

Keywords—E-commerce analytics, customer segmentation, RFM analysis, sales analytics, product analytics, feature engineering, data visualization, customer intelligence, Streamlit.

I. INTRODUCTION

A. Background

The rapid growth of e-commerce has resulted in the generation of large volumes of data related to customers, orders, products, payments, returns, and customer reviews. Such data contains valuable information about purchasing behaviour, product performance, sales trends, and revenue generation. However, analysing these data sources independently can make it difficult to obtain a comprehensive understanding of overall business performance.

Customer behaviour analysis is particularly important for identifying purchasing patterns, customer value, and differences among customer groups. Similarly, product-level and category-level analysis can help identify high-performing products, profitable categories, and products requiring further attention. Sales and revenue analysis can additionally reveal temporal and geographical trends that may support business planning and decision-making.

Traditional manual analysis of large e-commerce datasets is time-consuming and makes it difficult to continuously examine multiple business dimensions together. Therefore, an integrated analytical approach is required to preprocess heterogeneous e-commerce data, derive meaningful features, analyse customer and product behaviour, and present the resulting insights in an accessible form.

In this work, RetailIQ is developed as an end-to-end e-commerce analytics platform that combines data preprocessing, feature engineering, exploratory data analysis, customer intelligence, product intelligence, sales and revenue analysis, customer segmentation, and interactive visualization. The platform provides a centralized analytical view of different aspects of e-commerce operations and is intended to support systematic interpretation of business data and data-driven decision-making.

B. Problem Statement

E-commerce businesses generate data across multiple operational and customer-related sources, including customer profiles, orders, products, payments, returns, and reviews. When these data sources are analysed separately, it becomes difficult to obtain a unified understanding of customer behaviour and overall business performance. Raw and heterogeneous data also requires appropriate preprocessing before it can be effectively used for analysis.

In addition, identifying meaningful customer purchasing patterns, comparing product and category performance, analysing sales and revenue trends, and recognizing valuable customer segments can be challenging through manual analysis. The absence of a centralized analytical view can further increase the time and effort required to derive useful business insights.

Therefore, the problem addressed in this work is the development of an integrated e-commerce analytics platform capable of preprocessing and analysing multiple business data dimensions, identifying customer and product patterns, performing customer segmentation, and presenting the resulting insights through an interactive and centralized visualization interface.

C. Research Objectives

The main objective of RetailIQ is to develop an integrated e-commerce analytics platform that transforms raw e-

commerce data into meaningful customer, product, sales, and revenue insights. The specific objectives are:

1. To preprocess and integrate e-commerce data from customer, order, order-item, product, payment, return, and review sources for systematic analysis.
2. To analyze customer purchasing behavior and identify patterns based on transaction and customer attributes.
3. To analyze product and category performance using sales, revenue, profit, inventory, and customer-rating-related measures.
4. To analyze sales and revenue trends across time and geographical regions to identify important business patterns.
5. To perform customer segmentation using Recency, Frequency, and Monetary (RFM) analysis and clustering techniques.
6. To develop an interactive visualization dashboard that provides a centralized view of the generated analytical insights and key performance indicators.

D. Research Questions

The study is guided by the following research questions:

RQ1: How can heterogeneous e-commerce data be effectively preprocessed and integrated to support comprehensive business analysis?

RQ2: What customer purchasing patterns and customer segments can be identified from e-commerce transaction data using RFM analysis and clustering?

RQ3: How can product, category, sales, revenue, and geographical data be analyzed to identify meaningful business performance patterns?

RQ4: How can an interactive dashboard effectively present integrated e-commerce analytics and key business insights in a centralized interface?

E. Contributions

The main contributions of this work are as follows:
Integrated e-commerce analytics: RetailIQ combines customer, order, product, payment, return, and review data within a unified analytical workflow.
Comprehensive feature engineering: The platform derives temporal, customer, revenue, profit, discount, and inventory-related features to support downstream analysis.
Customer intelligence and segmentation: The system applies RFM analysis and clustering to identify distinct customer groups based on purchasing behavior and customer value.
Multi-dimensional business analysis: RetailIQ analyzes product and category performance, sales and revenue trends, geographical patterns, payment methods, returns, and customer reviews.
Interactive analytical dashboard: The generated insights are presented through an interactive dashboard, providing a centralized interface for exploring key performance indicators and analytical results.

F. Organisation of the Paper

The remainder of this paper is organized as follows. Section II presents the related work and discusses existing approaches for e-commerce analytics, customer segmentation, and related analytical techniques. Section III describes the proposed RetailIQ methodology, including the dataset, preprocessing, feature engineering, customer segmentation, analytical components, and dashboard architecture. Section IV presents the experimental setup, including the software and hardware environment, dataset preparation, model configuration, and evaluation metrics. Section V presents the experimental results and analytical findings. Section VI discusses the obtained results and their practical implications. Section VII describes the limitations of the proposed system. Finally, Section VIII concludes the paper and outlines directions for future work.

II. RELATED WORK

A. E-Commerce and Retail Analytics

Data-driven decision making has become foundational to modern retail operations. Early research by Ngai et al. [8] provided a comprehensive taxonomy of data mining techniques applied to Customer Relationship Management (CRM), emphasizing customer identification, attraction, retention, and development. Baesens et al. [15] highlighted the critical role of transactional data analysis in identifying purchasing patterns and optimizing promotional strategies. Subsequent studies by Birant [7] demonstrated that integrating customer transaction logs with product catalog attributes allows retailers to extract actionable patterns regarding consumer demand and inventory velocity. However, many early analytical frameworks focused primarily on offline brick-and-mortar transactional structures, requiring adaptation for the high-velocity, multi-channel nature of e-commerce environments.

B. Customer Behaviour and RFM Analysis

Recency, Frequency, and Monetary (RFM) analysis, originally formalized by Hughes [1], remains one of the most effective empirical models for evaluating customer value based on historical purchasing behavior. Chen et al. [6] combined RFM parameters with rough set theory to classify customer value tiers, demonstrating that behavioral recency and transaction frequency are strong predictors of future retention. Chen and Huang [13] extended RFM by discovering sequential purchasing patterns, proving that RFM metrics effectively reduce high-dimensional transaction data into meaningful behavioral vectors. Kumar and Prasad [10] applied RFM modeling to e-commerce customer retention, confirming that segmenting customers based on monetary contribution and purchasing recency enables targeted marketing strategies. RetailIQ builds upon these established RFM principles to group 22,820 active customers into distinct behavioral segments.

C. Machine Learning and Unsupervised Clustering

Unsupervised clustering algorithms, particularly K-Means proposed by MacQueen [2] and synthesized by Jain [16], are widely used to discover natural groupings in multidimensional customer datasets. Wei [4] demonstrated that coupling RFM-derived behavioral metrics with K-Means clustering significantly improves segmentation granularity compared to traditional rule-based cutoffs. Yu and Yuan [11] introduced improved K-Means implementations for e-commerce customer behavior, utilizing normalized feature vectors to mitigate scale distortion. To assess cluster validity, Rousseeuw [3] introduced the Silhouette Coefficient, providing a mathematical measure of cluster cohesion and separation. Kaufman and Rousseeuw [19] further established silhouette analysis as a standard internal validation index for determining optimal cluster count K. Alternative clustering paradigms—such as BIRCH [18], CURE [17], and hierarchical clustering—have also been studied, though K-Means remains favored in enterprise applications due to its interpretability and computational efficiency [12], [14]. In RetailIQ, K-Means is applied to normalized RFM feature space, and Silhouette analysis across K=2 to 7 is used to evaluate cluster quality.

D. Business Intelligence and Interactive Dashboards

Business Intelligence (BI) platforms and visual decision-support systems play a vital role in translating complex data analytics into actionable business insights [21]. Few [9] established core visual design principles for enterprise dashboards, emphasizing information density, visual hierarchy, and at-a-glance cognitive clarity. Chen and Zhang [22] explored visual analytics in e-commerce decision-support systems, showing that interactive visual filtering enables managers to quickly isolate performance anomalies across products and regions. Stone [23] demonstrated that interactive dashboard architectures foster faster executive alignment and data-informed decision-making. In modern data science workflows, Python-based frameworks such as Streamlit, Plotly, and Pandas [30] have emerged as powerful tools for developing responsive, open-source BI web applications [26]. RetailIQ leverages these tools to construct an enterprise-grade multi-page dashboard interface.

E. Critical Comparison

The reviewed approaches differ in their objectives, data representations, and levels of automation. Topic-modeling approaches are useful for discovering recurring themes in complaint data, but they do not directly provide detailed complaint classification or operational routing. Traditional machine-learning methods provide computationally efficient classification and prediction capabilities but generally depend on manually designed or frequency-based text representations. Deep-learning and transformer-based approaches provide stronger contextual representations and have been applied to complaint and banking intent classification, but their focus remains primarily on prediction rather than end-to-end complaint management. Other approaches address specific aspects such as emotion or anomaly detection without integrating these capabilities into a unified complaint-management workflow. Overall, the existing approaches demonstrate the effectiveness of automated text analysis but leave a gap between complaint understanding and subsequent management actions such as prioritization, departmental routing, ticket generation, employee assignment, and tracking.

F. Research Gap and Comparative Summary

While existing literature extensively addresses individual analytical domains—such as standalone RFM segmentation [1], [6], K-Means customer clustering [4], [11], or visual dashboard design [9], [22]—there remains a distinct lack of research presenting a fully integrated end-to-end e-commerce analytics framework. Most existing studies isolate customer analytics from operational return drivers, product-level profit margin analysis, and regional geographical performance. RetailIQ bridges this research gap by providing an end-to-end framework that integrates multi-source preprocessing, comprehensive feature engineering, dual customer segmentation (RFM and K-Means), category profitability analysis, operational return tracking, sentiment processing, and interactive multi-page dashboard visualization within a single unified workflow.

III. PROPOSED METHODOLOGY

A. Overall Framework Architecture

The RetailIQ framework follows a systematic, six-stage analytical workflow designed to transform heterogeneous raw e-commerce data into structured strategic business intelligence. As illustrated in the pipeline architecture, the workflow proceeds through: (1) Relational Data Ingestion, (2) Data Inspection & Preprocessing, (3) Schema Integration & Merging, (4) Cross-Domain Feature Engineering, (5) Multi-Dimensional Analytics Modules (RFM Segmentation, K-Means Clustering, Product/Category Analytics, Regional Geography, Operations, Reviews), and (6) Interactive BI Dashboard Presentation.

B. Dataset Description

This study utilizes a comprehensive simulated Indian e-commerce dataset comprising seven relational tables representing a realistic multi-category online retail ecosystem. The dataset contains 25,000 unique customer profiles, 100,000 transaction orders, 254,331 granular order-item records, 550 products spanning 22 categories, 100,000 payment transaction records, 12,075 product return logs, and 77,530 customer review records. Table I summarizes the dataset schema and record counts.

Dataset Name	Record Count	Core Operational Attributes
customers.csv	25,000	Age, Gender, State, City, Signup Date, Loyalty Tier
orders.csv	100,000	Customer ID, Order Date, Order Status, Coupon, Campaign
order_items.csv	254,331	Order ID, Product ID, Quantity, Unit Price, Revenue, Profit
products.csv	550	Category, Subcategory,

		Brand, Cost Price, Stock, Rating
payments.csv	100,000	Order ID, Payment Method, Payment Status, Amount Paid
returns.csv	12,075	Order ID, Return Date, Return Reason, Refund Status
reviews.csv	77,530	Product ID, Customer ID, Rating, Review Text, Verified

C. Data Preprocessing & Quality Analyis

Data cleaning and standardization were systematically applied across all seven input datasets to resolve missing values, correct data type mismatches, and ensure referential integrity:

• Temporal Standardization: All date fields across customers, orders, returns, and reviews were converted from text strings to standardized ISO datetime objects (YYYY-MM-DD HH: MM: SS).

• Numerical Missing Imputation: Missing customer age values were imputed using the dataset median age (37.0 years). Missing product rating values were imputed using the category median rating.

• Categorical Missing Treatment: Missing preferred device values in customer profiles were categorized as 'Unknown'. Missing coupon codes and promotional campaign IDs in orders were filled with 'No Coupon' and 'No Campaign', respectively.

• Review Sentiment Derivation: For review records missing explicit text sentiment labels, sentiment was derived deterministically from the numerical star rating: ratings of 4 or 5 were classified as 'Positive', 3 as 'Neutral', and 1 or 2 as 'Negative'.

• Preservation of Structural Missingness: Missing last order dates for inactive customers with zero purchase history were intentionally retained as null rather than imputed, accurately preserving non-purchasing behavioral states.

D. Cross-Domain Feature Engineering

Feature engineering was conducted on the integrated dataset to derive critical financial, temporal, inventory, and customer behavioral attributes:

• Temporal Features: Extracted order_year, order_month, month_name, order_quarter, order_day, and day_name from order dates to analyze seasonality and day-of-week sales velocity.

• Financial & Margin Metrics: Calculated item_profit = revenue - cost, profit_margin_pct = (item_profit / revenue) * 100, and profit_per_unit = item_profit / quantity to evaluate unit economics.

• Discount Metrics: Derived discount_amount = (unit_price * quantity * discount_pct) / 100 to measure gross promotional margin erosion.

• Customer Tenure at Order Time: Engineered tenure_at_order_days = (order_date - signup_date). days, representing the exact customer account age at the time of purchase.

• Inventory Valuation: Calculated inventory_value = stock_quantity * cost_price to assess working capital tied up in warehouse stock.

• Binary Return Mapping**:** Engineered has_return = 1 if the order_id exists in returns.csv, else 0, mapping 25,739 order-item records (across 9,510 unique orders) to return events without introducing one-to-many join duplication.

E. Customer Analysis & RFM Segmentation Methodology

Customer purchasing behavior was modeled using Recency, Frequency, and Monetary (RFM) analysis. For each of the 22,820 customers with recorded transaction activity, three core metrics were computed relative to the dataset snapshot reference date:

• Recency (R): Days elapsed between the customer's most recent order date and the reference snapshot date.

• Frequency (F): Total count of unique completed orders placed by the customer.

• Monetary (M): Total net monetary spend cumulative across all completed transaction orders.

Quintile scoring (1 to 5) was applied to assign R, F, and M scores. Customers were subsequently segmented into six business-actionable behavioral tiers based on rule-based RFM score combinations: Champions (R=4-5, F=4-5, M=4-5), Loyal Customers (F=3-5, M=3-5), New Customers (R=4-5, F=1-2), At Risk (R=1-2, F=3-5, M=3-5), High Value At Risk (R=1-2, F=4-5, M=4-5), and Regular Customers (all remaining active profiles).

F. Unsupervised K-Means Clustering Methodology

To complement rule-based RFM segmentation with data-driven unsupervised groupings, K-Means clustering was executed on the log-transformed and standardized customer RFM metrics. Features (R, F, M) were log-transformed to eliminate heavy right-skewness and normalized using StandardScaler (zero mean, unit variance). K-Means models were trained for cluster counts K = 2 through 7 using random_state = 42 and n_init = 10. Cluster evaluation was performed using the Silhouette Coefficient (s):

$$s(i) = [\, b(i) - a(i) \,] / \max(\, a(i), b(i) \,)$$

where a(i) represents the mean intra-cluster distance of sample i, and b(i) represents the mean nearest-cluster distance of sample i. The mean silhouette score across all samples was computed to determine the mathematical optimal K.

IV. EXPERIMENTAL SETUP

A. Software and Execution Environment

The RetailIQ analytical pipeline and interactive dashboard were implemented using Python 3.13 within a 64-bit Windows software environment. Key software libraries and version configurations include:

• Data Processing & Numerical Computing: Pandas 2.2.3 and NumPy 2.1.2 for relational data manipulation, aggregation, and feature transformation.

• Machine Learning & Statistical Validation: Scikit-Learn 1.6.1 for StandardScaler feature normalization, K-Means clustering execution, and silhouette coefficient score calculation.

• Data Visualization & Plotting: Plotly 6.0.0 and Matplotlib 3.10.0 for generating high-resolution interactive charts, geographic choropleth maps, and statistical plots.
• Business Intelligence Web Dashboard: Streamlit 1.42.0 for building and serving the multi-page web dashboard application.

B. Hardware Environment

Experiments were conducted on a standard workstation equipped with an x86_64 Multi-Core Processor running at 2.50 GHz, 16 GB RAM, and solid-state storage. All feature engineering and clustering tasks executed locally within reasonable computation times.

C. Analytical & Model Configuration

The unsupervised K-Means clustering pipeline was configured with strict reproducible parameters: feature scaling via StandardScaler, convergence tolerance of 1e-4, maximum iterations set to 300, random state seed locked at 42, and n_init set to 10 initializations. K values from 2 through 7 were evaluated sequentially.

D. Evaluation Metrics

Model and system evaluation relies on standard mathematical metrics:
• Silhouette Coefficient (s): Evaluates clustering compactness and separation in range [-1, +1].
• Financial Profit Margin (%): Profit Margin = (Total Profit / Total Revenue) * 100.
• Return Rate (%): Return Rate = (Total Orders Returned / Total Completed Orders) * 100.
• Review Sentiment Ratio (%): Percentage distribution of Positive, Neutral, and Negative reviews.

V. EXPERIMENTAL RESULTS AND ANALYTICAL FINDINGS

A. Data Integration and Preprocessing Results

The data integration pipeline successfully joined 100,000 orders with 254,331 order-items, 25,000 customers, and 550 products into a unified master analytical dataset comprising 254,331 records and 36 engineered attributes. Preprocessing resolved all missing customer ages (median age 37.0) and device types (assigned 'Unknown' to missing profiles), establishing 100% referential integrity across all downstream analytical modules.

B. Customer RFM Segmentation Results

RFM analysis was successfully conducted on 22,820 active customers who placed at least one order. Applying rule-based quintile scoring categorized the customer base into six distinct behavioral tiers. Table II reports the exact customer distribution and behavioral characteristics across RFM segments.

RFM Segment Name	Customer Count	Share (%)	Avg Recency (Days)	Avg Frequency	Avg Spend (₹)
Regular Customers	5,693	24.95 %	396.4	2.8	51,240
Loyal Customers	5,628	24.66 %	301.2	8.9	271,850
At Risk	4,945	21.67 %	412.5	3.1	58,910
Champions	3,119	13.67 %	145.8	11.4	385,420
New Customers	2,967	13.00 %	98.2	1.4	22,150
High Value At Risk	468	2.05 %	448.1	9.2	295,640

Table II: RFM customer segmentation distribution & behavioral profile

C. K-Means Clustering & Silhouette Evaluation Results

Unsupervised K-Means clustering was executed across K values from 2 to 7 on normalized RFM vectors. Table III presents the empirically calculated Silhouette Coefficients across all evaluated cluster counts.

Cluster Count (K)	Silhouette Coefficient	Clustering Structure Quality
K = 2	0.4378	Optimal Cohesion & Separation
K = 3	0.4107	Strong Separation
K = 4	0.3983	Moderate Separation
K = 5	0.4013	Moderate Separation
K = 6	0.3338	Sub-optimal Overlap
K = 7	0.3419	Sub-optimal Overlap

Table III: K-Means Clustering Silhouette Evaluation (K=2 To 7)

The empirical evaluation confirms that K=2 achieves the highest silhouette score (0.4378), effectively bisecting the customer base into Cluster 0: Core Moderate Shoppers (17,284 customers; avg recency 387.7 days, avg frequency 2.9 orders, avg spend ₹53,416.89) and Cluster 1: High-Value Frequent Shoppers (5,536 customers; avg recency 295.0 days, avg frequency 9.01 orders, avg spend ₹275,766.17).

D. Product & Category Profitability Results

Analysis of product category sales revealed striking discrepancies between top-line revenue generation and net profit margins. Electronics generated the highest gross revenue (₹1,223,618,824.25 across 59,068 items), but yielded an exceptionally low profit margin of only 0.69% (net profit ₹8,450,905.95). In contrast, Fashion generated ₹139,393,560.30 in revenue across 98,750 items, but achieved a substantial profit margin of 33.14% (net profit ₹46,192,014.22). Home & Kitchen contributed ₹158,075,531.70 in revenue with a 17.96% margin (₹28,392,024.89 profit). Table IV summarizes performance across key product categories.

Category Name	**Items Sold**	**Total Revenue (₹)**	**Total Profit (₹)**	**Profit Margin**	**Avg Rating**
Electronics	59,068	1,223,618,824.25	8,450,905.95	0.69 %	3.97
Appliances	35,842	405,233,265.45	15,743,436.40	3.89 %	4.00
Home & Kitchen	60,205	158,075,531.70	28,392,024.89	17.96 %	3.92

Furniture	8,118	147,011,493.10	14,348,527.85	9.76%	3.87
Fashion	98,750	139,393,560.30	46,192,014.22	33.14%	4.01
Footwear	32,501	82,603,494.90	24,298,429.50	29.42%	4.04
Bags	29,948	55,912,111.45	15,984,391.42	28.59%	4.20
Health & Wellness	36,095	31,824,329.15	6,957,754.85	21.86%	4.21

Table IV: Key Product Category Sales, Revenue, And Margin Performance

Furthermore, granular item-level auditing identified exactly 33 individual SKUs operating at a negative net profit margin, representing loss-making products requiring immediate pricing or sourcing renegotiation.

E. Geographical Analysis Results

Regional geographical performance was analyzed across all Indian states and major urban centers within the dataset. Maharashtra, Karnataka, Tamil Nadu, Delhi, and Uttar Pradesh emerged as the top revenue-generating territories. State-level profit margins remained consistent between 11.2% and 13.5% across major regions, reflecting balanced nationwide distribution within the simulated dataset.

F. Operations: Returns and Payment Method Analysis

Operational auditing of transaction orders revealed a total of 12,075 returns out of 100,000 completed orders, establishing an overall baseline Return Rate of 12.07%. Analysis of return logs identified the top return drivers: 'Size issue' (2,201 returns; 18.2%), 'Changed mind' (1,984 returns; 16.4%), 'Product damaged' (1,933 returns; 16.0%), 'Quality issue' (1,692 returns; 14.0%), and 'Product not as expected' (1,442 returns; 11.9%).

Payment channel analysis indicated that Unified Payments Interface (UPI) is the primary payment method, capturing 31,971 transactions (31.97% share, ₹780,563,148.81 total volume). Cash on Delivery (COD) ranked second with 17,274 transactions (₹398,801,556.21 volume), followed by Credit Card (15,063 transactions) and Debit Card (12,682 transactions).

G. Customer Review & Sentiment Results

Analysis of 77,530 customer review records revealed an overwhelmingly positive sentiment distribution: 46,311 reviews (59.73%) were Positive (ratings 4-5), 24,208 reviews (31.22%) were Neutral (rating 3), and 7,011 reviews (9.04%) were Negative (ratings 1-2). Product categories with highest average satisfaction included Pet Supplies (4.40 avg rating), Toys (4.36), and Health & Wellness (4.21).

H. Multi-Page Interactive Dashboard Results

The analytical outputs were fully integrated into a multi-page interactive Streamlit business intelligence dashboard. The application provides centralized navigation across 7 specialized analytical modules:

1. Executive Overview: High-level summary KPIs (Gross Revenue, Total Profit, Active Customers, Total Orders, Overall Return Rate) and executive performance cards.
2. Sales Analytics: Temporal sales velocity charts, monthly revenue trends, order volume distribution, and discount erosion analysis.
3. Customer Intelligence: RFM segment distribution charts, K-Means silhouette cluster plots, customer tenure metrics, and segment strategy recommendations.
4. Product Intelligence: Category revenue vs margin scatter matrices, subcategory performance tables, and loss-making product audit lists.
5. India Analytics: State-level choropleth revenue maps, regional profit breakdowns, and city-level performance rankings.
6. Returns & Payments: Return reason distribution bar charts, return rate metrics, and payment gateway method breakdowns.
7. Reviews & Customer Experience: Rating distributions, sentiment breakdown charts, verified purchase filters, and category rating leaderboards.

VI. DISCUSSION

A. Interpretation of Analytical Findings

The experimental findings provide critical insights into e-commerce operations. The RFM customer segmentation demonstrates that customer value is highly concentrated: Champions (13.67%) and Loyal Customers (24.66%) account for over 65% of net enterprise profitability despite comprising under 39% of the customer base. Conversely, 21.67% of customers are 'At Risk' and 2.05% are 'High Value at Risk', representing significant revenue vulnerable to churn if targeted re-engagement campaigns are not deployed.

Product performance analysis highlights the danger of relying solely on top-line revenue metrics. Electronics generates over ₹1.22 billion in revenue but yields a meager 0.69% profit margin due to high cost price and aggressive discounting. Conversely, Fashion yields ₹46.19 million in net profit at a 33.14% margin. Identifying 33 loss-making products empowers retail merchandisers to adjust prices or prune unprofitable inventory.

Operational return analysis demonstrates that size mismatch (18.2%) and damaged goods (16.0%) account for over one-third of all returns, pointing directly to actionable operational remedies such as improved sizing charts and enhanced protective packaging.

B. Enterprise Stakeholder Decision Support

RetailIQ provides customized decision support across key enterprise roles:

• Executive Leadership: Monitors overall enterprise health, net profitability, and high-level return rates via the Executive Overview module.
• Marketing & CRM Teams: Leverages RFM and K-Means customer segments to design tailored retention campaigns for 'High Value At Risk' customers.
• Merchandising & Product Managers: Isolates loss-making SKUs and shifts promotional focus toward high-margin categories like Fashion and Footwear.
• Operations & Logistics Directors: Tracks return drivers and optimizes fulfillment processes to mitigate product damage and size mismatches.
• Financial Analysts: Monitors payment gateway adoption, UPI transaction shares, and discount erosion metrics.

VII. LIMITATIONS

To maintain academic rigor, several methodological limitations of this study must be acknowledged:

1. Simulated Dataset Boundaries: The study relies on a simulated Indian e-commerce dataset. While statistically robust, empirical findings (e.g., category margins, return rates) reflect the underlying simulation distributions and cannot be interpreted as universal market truths for specific real-world platforms like Amazon or Flipkart.
2. Static Batch Processing: The current architecture operates on a fixed dataset snapshot rather than a live streaming data pipe, requiring periodic script re-execution for updated dashboard refreshes.
3. External Validity: Geographical and category insights derived from this dataset cannot be automatically generalized to international markets or niche specialized verticals.
4. Unsupervised Model Sensitivity: K-Means clustering outputs depend on the selected feature scaling and cluster count K, requiring periodic re-validation as customer purchasing behaviors evolve.

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented RetailIQ, an integrated end-to-end e-commerce analytics framework and interactive business intelligence dashboard. By ingesting, cleaning, and integrating heterogeneous transactional data across 25,000 customers, 100,000 orders, 254,331 order-items, 550 products, 100,000 payments, 12,075 returns, and 77,530 reviews, the platform establishes a seamless analytical pipeline. Cross-domain feature engineering derived temporal, financial, customer tenure, item margin, and return attributes. RFM modeling segmented 22,820 active customers into 6 behavioral tiers, while K-Means clustering evaluated cluster quality across K=2 to 7 via Silhouette Coefficients (identifying optimal K=2 with s=0.4378). Comprehensive product profitability auditing identified 33 loss-making SKUs, operational analysis uncovered a 12.07% return baseline driven by size and damage issues, and sentiment analysis confirmed 59.73% positive feedback. All insights were deployed within a multi-page interactive Streamlit dashboard, demonstrating how unified analytics empowers enterprise stakeholders to execute data-driven retail decisions.

B. Future Work

Future extensions of the RetailIQ framework will focus on:

• Automated Streaming Ingestion: Integrating Apache Kafka or Airflow pipelines for live real-time transaction ingestion.
• Database & API Integration: Connecting the analytics pipeline directly to PostgreSQL/Snowflake enterprise data warehouses.
• Predictive Customer Analytics: Developing supervised machine learning models for customer churn prediction and Customer Lifetime Value (CLV) forecasting.
• Advanced NLP Sentiment Extraction: Implementing transformer-based models (e.g., BERT) for fine-grained aspect-based review sentiment mining.
• Cloud Infrastructure Deployment: Containerizing the Streamlit BI application with Docker and deploying on AWS/GCP cloud environments.

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